Threshold Bids and the Concavity Requirement

Where the MCP work has got to, and a proposed experiment

Discussion note · September 2026

1

Where we are

The MCP, the equilibrium result, and what is still open

The equilibrium result, and what it rests on

ESTABLISHED

- The compact source model expanded, every constraint traced to its origin
- A multi-agent model stated independently, as an MCP over a box
- KKT(ISO) is partitioned exactly by the agents' systems, each condition used once
- Under the assumptions, a solution is a competitive equilibrium
- Individual rationality and revenue adequacy follow as corollaries

THE ASSUMPTIONS IT NEEDS

- Price taking: standard, and implicit in the paper
- Convex market: no binaries, affine constraints; also in the paper
- Separability by ownership: needed to split the constraints at all
- Bid validity: w(e) non-increasing, i.e. b(e) concave: NOT in the paper, and the one we want to discuss

Separability buys the partition; convexity buys the economics. Neither implies the other, which is why they are separate hypotheses.

The one assumption that is not innocuous

Bid validity is what makes the storage owner's KKT conditions sufficient — what lets us say the owner is maximising rather than that a derivative vanishes. So the equilibrium claim depends on it entirely.

- For a load, requiring a non-increasing bid costs nothing: the dispatch would reorder the blocks anyway, so the monotone stack is equivalent.
- For storage it is not equivalent. w(e) values a stock, and a stock can have increasing returns, a threshold benefit is real and ordinary.
- Such an owner cannot declare its true valuation at all. Not “would rather not”, cannot: the format has no admissible representation for it.

Proposed: measure what that costs. Is it worth the time?

2

Why bids are not always non-increasing

A threshold in the stock breaks concavity

A discrete benefit at a stock level is a spike in the margin

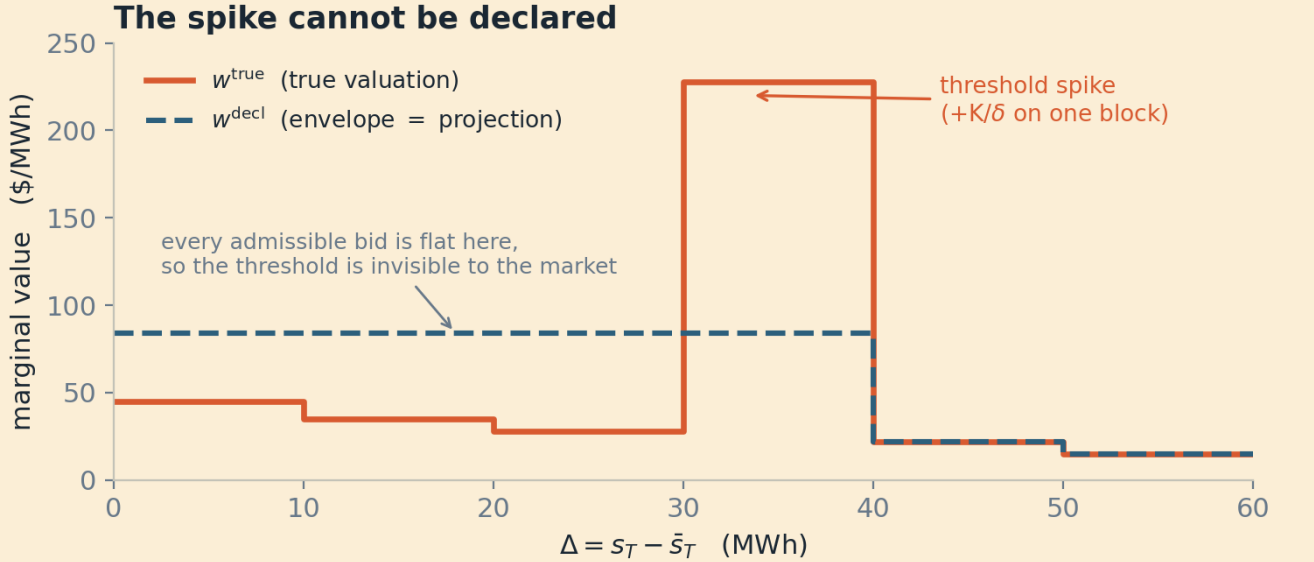
THE OWNER'S TRUE VALUATION

$$b^{\text{true}}(\Delta) = \underbrace{\sum_{j=1}^{J} v_j \, \Delta_j}_{\text{arbitrage (concave)}} + \underbrace{K \cdot \mathbf{1}\{\Delta \geq \Delta_0\}}_{\text{threshold (not concave)}}, \qquad \Delta = \sum_j \Delta_j, \quad 0 \leq \Delta_j \leq \delta$$

$$w_j^{\text{true}} = v_j + \frac{K}{\delta} \mathbf{1}\{\text{block } j \text{ ends at } \Delta_0\} \quad \Longrightarrow \quad w^{\text{true}} \text{ increases at one block}$$

- The arbitrage part is concave, successive MWh go into successively worse hours of the next day. That part behaves.
- The threshold term is an indicator. A jump in total value is, by construction, a spike in marginal value and a spike is an increase.
- b(true) is therefore non-concave for any K > 0, however smooth the arbitrage is. This is formally a start-up cost.

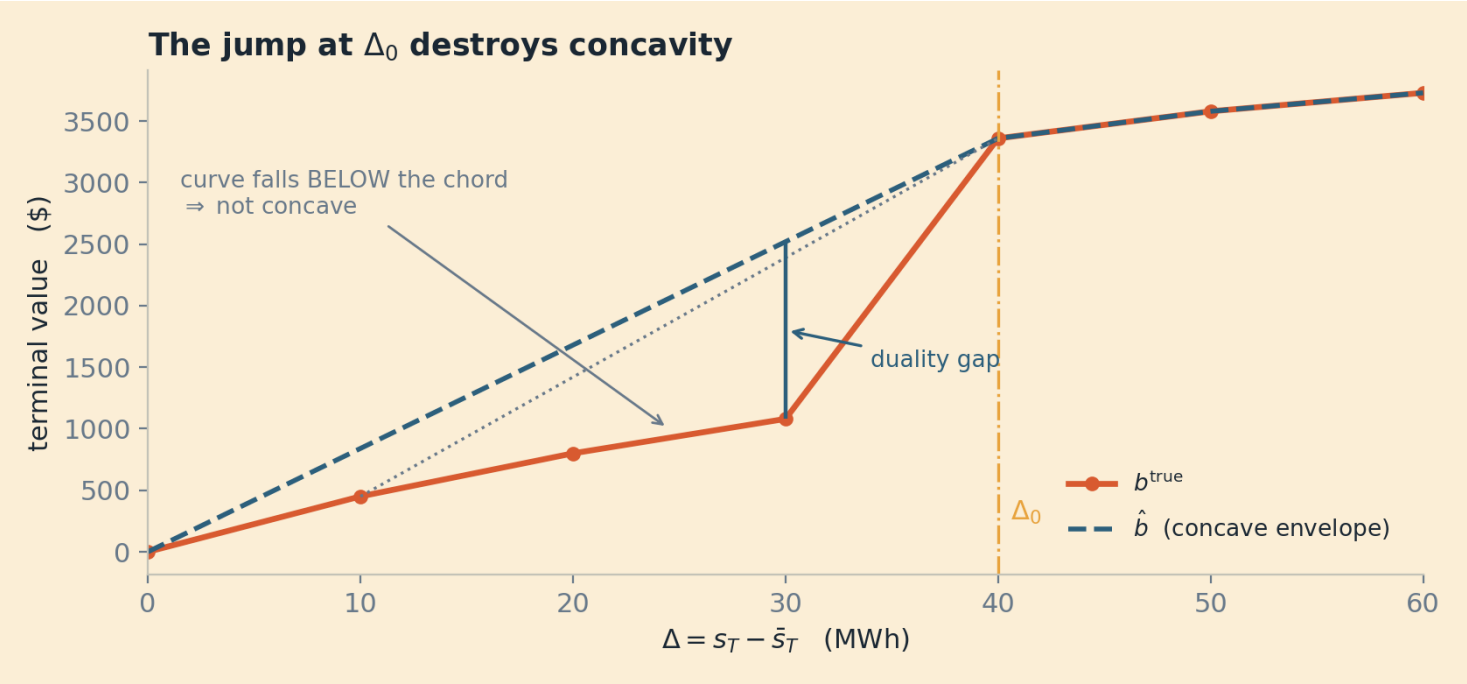
The spike cannot be declared



Both flattening rules give the same answer: the slopes of the concave envelope are exactly the isotonic projection.

Both flattening rules coincide, and that is a theorem: the slopes of the least concave majorant are exactly the isotonic projection. There is no third option.

The jump destroys concavity



The vertical distance between b(true) and its concave envelope is the duality gap — the same quantity that justifies uplift payments in unit commitment.

Optimal against the bid, inefficient against the truth

1 · The bid is flattened

Every admissible bid is flat over the run-up. The information “these MWh are critical to me” is gone.

2 · The ISO clears correctly

π lands anywhere in the flat region and (KA8) is satisfied. The dispatch is optimal for the bid received.

3 · The threshold is missed

The owner ends below Δ_o and pays the penalty. Welfare is lower than it needed to be.

The owner never misreports. It declares the closest admissible shape to what it believes, and the total value it declares is unchanged — only redistributed. The loss is attributable to the expressive power of the bid format, not to strategy.

Truthfulness asks whether a participant WANTS to report truthfully. This asks whether it CAN — and that question is logically prior.

3

This happens in practice

Threshold rules in ERCOT and CAISO

A rule written directly on state of charge

- Since the December 2022 Business Practice Manual update, an ESR must be able to satisfy its discharging AS awards for the full duration of the product.
- 10 MW of RegUp or RRS (1-hour products) ⇒ at least 10 MWh of SOC at the start of the hour. 10 MW of ECRS, which had a 2-hour requirement ⇒ 20 MWh.
- Tightened further by NPRR1186 (approved April 2024), and under RTC+B each MW of Non-Spin requires 4 MWh stored at the start of each 5-minute interval.

This is exactly Δ_o. At 20 MWh the full ECRS award is kept; at 19.9 MWh it is not. The MWh that crosses the threshold carries the value of the whole award.

- And it bites: operators objected publicly that the rule forced them to withhold ancillary capacity during peak demand.

Resource adequacy credit at four hours

- The CPUC sets a battery's net qualifying capacity from tested sustained output over a four-hour period.
- Batteries with longer discharge duration receive no more RA credit than those that can only manage four hours.

The threshold

Below 400 MWh (for 100 MW) the RA credit falls; on crossing it the whole block is accredited. CAISO's own material states the equivalence: 400 MWh = 100 MW for 4 hours.

And the flat part after it

Past four hours the marginal MWh is worth nothing for RA. So the true valuation is high-flat, jump, low-flat — manifestly non-concave.

CAISO's 2023 battery storage report also notes that on high-priced days part of the fleet discharges early and cannot recharge before the critical peak — the failure this experiment predicts, already observed.

4

The proposed experiment

The model, and what we expect to find

One bus, one battery, parameterised by the terminal bid

P(Φ) — MODEL A WITH THE NETWORK DELETED

$$
\begin{aligned}
\mathrm{P}(\Phi): \quad \min_{q,r,p,g,s} \quad & \Delta T \sum_{t\in\mathcal{T}} \Big[\sum_{k} c_k q_{k,t} + \mathrm{VOLL}\, r_t + \epsilon p_t + \zeta g_t \Big] \; - \; \Phi(\Delta) \\
\text{s.t.} \quad & \sum_{k} q_{k,t} + g_t - p_t + r_t - D_t = 0 && [\ \Delta T \lambda_t\ ] \quad \text{(E1)} \\
& 0 \le q_{k,t} \le \bar{q}_k, \qquad r_t \ge 0 && \text{box} \quad \text{(E2)} \\
& s_t - \gamma s_{t-1} - \Delta T \alpha p_t + \Delta T \beta g_t = 0 && [\ \pi_t\ ] \quad \text{(E3)} \\
& \gamma s_{t-1} + \Delta T \alpha p_t \le \bar{s}, \quad \underline{s} \le \gamma s_{t-1} - \Delta T \beta g_t && [\ \bar{\nu}_t,\ \underline{\nu}_t\ ] \quad \text{(E4)} \\
& 0 \le p_t \le \bar{p}, \qquad 0 \le g_t \le \bar{g} && \text{box} \quad \text{(E5)}
\end{aligned}
$$

Single bus, T = 4, no ramps, inelastic demand plus a shedding block at VOLL for feasibility. Φ is whichever terminal value function the case uses.

What the system would get if the truth were declarable

SOLVED EXACTLY BY TWO LINEAR PROGRAMS

$$W_{\mathrm{A}} = \max_{\Delta}\Big\{\, W^{\star}(\Delta) + K \cdot \mathbf{1}\{\Delta \geq \Delta_0\} \,\Big\} \qquad \text{non-convex}$$

all the non-convexity is a single binary, so:

$$W_0 = \text{ value of P(arbitrage only)} \qquad \text{branch } z = 0 \;\; \text{(one LP)}$$

$$W_1 = \text{ value of P(arbitrage only) with } \Delta \geq \Delta_0, \text{ plus } K \qquad \text{branch } z = 1 \;\; \text{(one LP)}$$

$$\boxed{W_{\mathrm{A}} = \max\{W_0,\, W_1\}} \qquad \text{exact, not a relaxation}$$

- Not a market outcome — a reference ceiling. The bid validator would reject w(true) outright.
- Case A carries no λ: with a discrete decision embedded, no uniform linear price supports the dispatch. Same obstruction as unit commitment uplift.

Always scored against the true valuation

EVALUATED WITH b(true) IN EVERY CASE

$$\text{system welfare} \;=\; -\Delta T \sum_t \Big[\sum_k c_k q_{k,t} + \text{VOLL}\, r_t + \epsilon p_t + \zeta g_t \Big] \;+\; b^{\text{true}}(\Delta)$$

$$\text{owner surplus} \;=\; \Delta T \sum_t \big[\lambda_t (g_t - p_t) - \epsilon p_t - \zeta g_t \big] \;+\; b^{\text{true}}(\Delta)$$

Scoring each case against its own declared bid would compare every run to its own yardstick and reveal nothing. This is the crux of the design.

- Also reported: Δ* and whether the threshold is met; the λ and π paths for B, C, D; the duality gap; and the penalty incurred.
- Ten validation checks, two of which are the correspondence conditions (KB5)–(KB7) — so passing them confirms the code matches the theory already written up.

The shape of the result, not yet the numbers

case	bid used	Δ*	threshold	welfare	note
A	true valuation, two branches	≥ Δ_o	met	highest	n/a
B	concave envelope	< Δ_o	missed	lower	= C
C	isotonic projection	< Δ_o	missed	lower	= B
D	control, K = 0	same for all three	n/a	identical	gap = 0

The number of interest is the welfare gap between A and B/C. It is the price of a bid format that cannot represent increasing returns in the stock — and the owner is sincere in every run.

Calibration target: with K = 0 the optimum should land just short of Δ_o, so the threshold changes the decision without the case looking tuned.